II. a. Questions

1) Explain the relationship between sampling rate and frequency response mathematically.

by the **Nyquist theorem** Sampling rate (F_s) must be at least twice the max frequency (f_{max}) of the signal.

$F_s \geq 2 \cdot f_{max}$

2) If we record 10 seconds of sound with 16-bit resolution stereo with sapling rate of 44100 Hz, calculate the size of the audio file.

↳ 2 byte

channel * byte * Sapling rate * second = $2 \cdot 2 \cdot 44100 \cdot 10 = 1{,}764{,}000$ Byte

1.764 MB

Stereo is two channel (L, R) = 2 channel

16 bit resolution => 16/8 = 2 byte

3) Assume that a sample of audio signal is 0.501. Represent this value using 16 and 8 bit resolution for binary floating point representation.

0,501 . 2 = 1,002	=>	1
0,002 . 2 = 0,004	=>	0
0,004 . 2 = 0,008	=>	0
0,008 . 2 = 0,016	=>	0
0,016 . 2 = 0,032	=>	0
0,032 . 2 = 0,064	=>	0
0,064 . 2 = 0,128	=>	0
0,128 . 2 = 0,256	=>	0
0,256 . 2 = 0,512	=>	0
0,512 . 2 = 1,024	=>	1
0,024 . 2 = 0,048	=>	0
0,048 . 2 = 0,096	=>	0
0,096 . 2 = 0,192	=>	0
0,192 . 2 = 0,384	=>	0
0,384 . 2 = 0,768	=>	0
0,768 . 2 = 1,536	=>	1

→ 8bit

→ 16bit

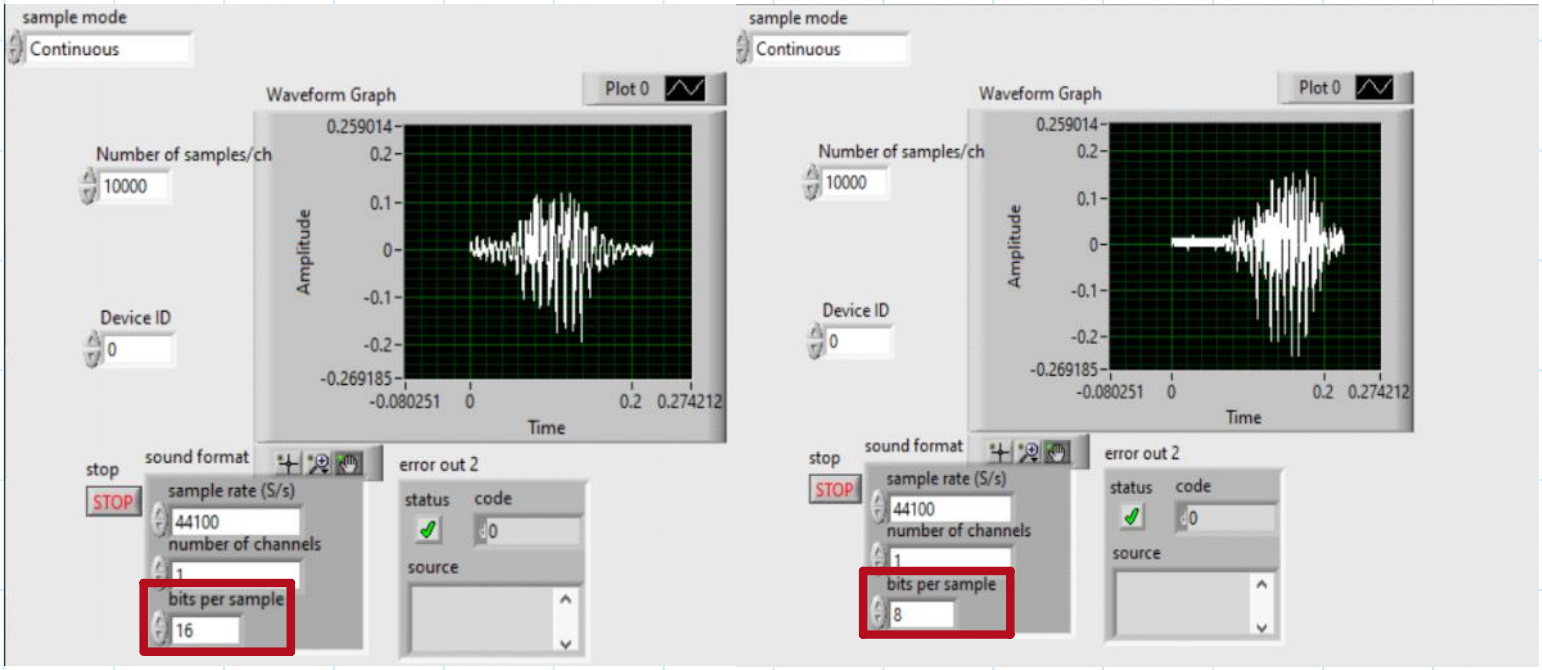
10000000 → 8 bit

1000000010000001 → 16bit

1. Assume we want to record the sound with the above parameters for 10 sec. Calculate the size of the audio file.

1) If we assume we only record a audio file for 10 seconds with parameters, Sample Rate 44100, 16-bit, with one channel.
We should do calculate the formula SampleRate*Byte*Channel*Seconds = AudioSize in Byte.
So we have 16/8 = 2 Byte, and if we apply the formula of AudioSize, we get 44100*2*1*10=882,000 Byte Of audio size.

2. Reduce the bit resolution from 16 bits to 8 bits. See the difference by zooming on the signal and explain the effect in your report.



2) When we reduce bits per sample to 8, we actually effectively reducing the precision with which each sample can be represented.
Amplitude of the signal is decreasing while we are lowering the bit count. By that soft and loud sounds may be less understandable

3. Explain what is number of samples/ch and how it effects on the audio signal.

3) the number of samples per channel affects the accuracy and quality of the audio signal. When we try to increase or decrease the number of samples, program starts lagging. Finding appropriate number of samples is crucial for analyzing

4. Calculate dt, the time interval between samples by using sample rate of sound format.

   Note that you need to unbundle sound format cluster. ($dt = 1 / \text{sample rate}$).

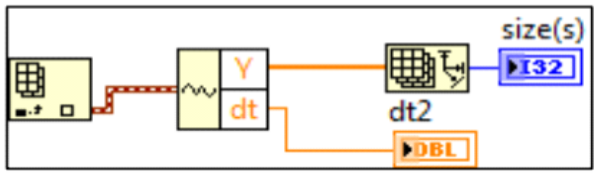
dt = 1/SampleRate(44100) dt = 1/44100

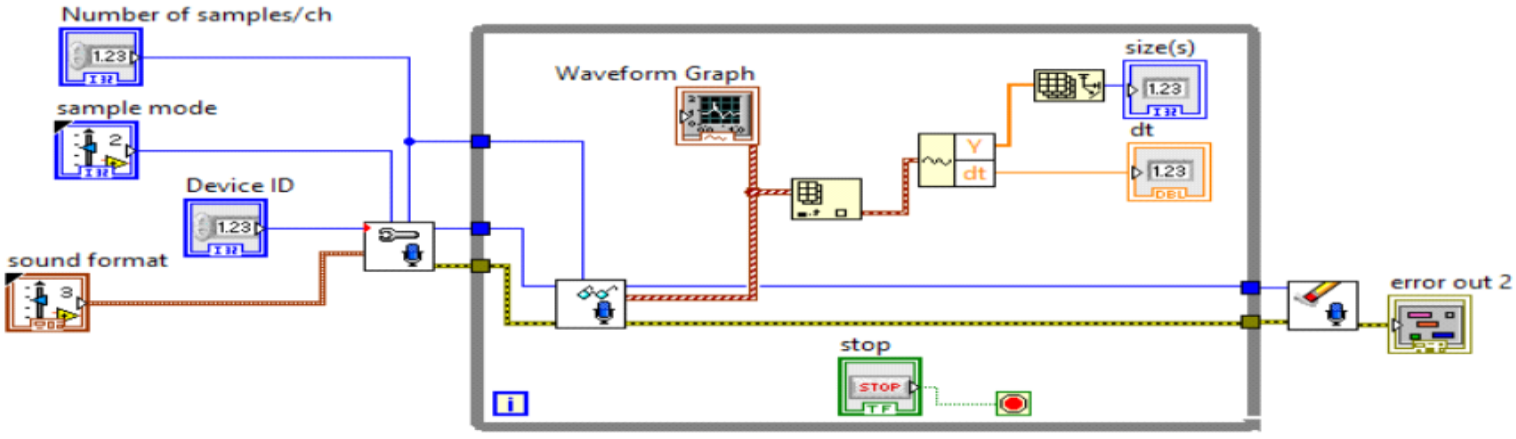
5. Calculate the time of each fetch by using dt and number of samples/ch. (

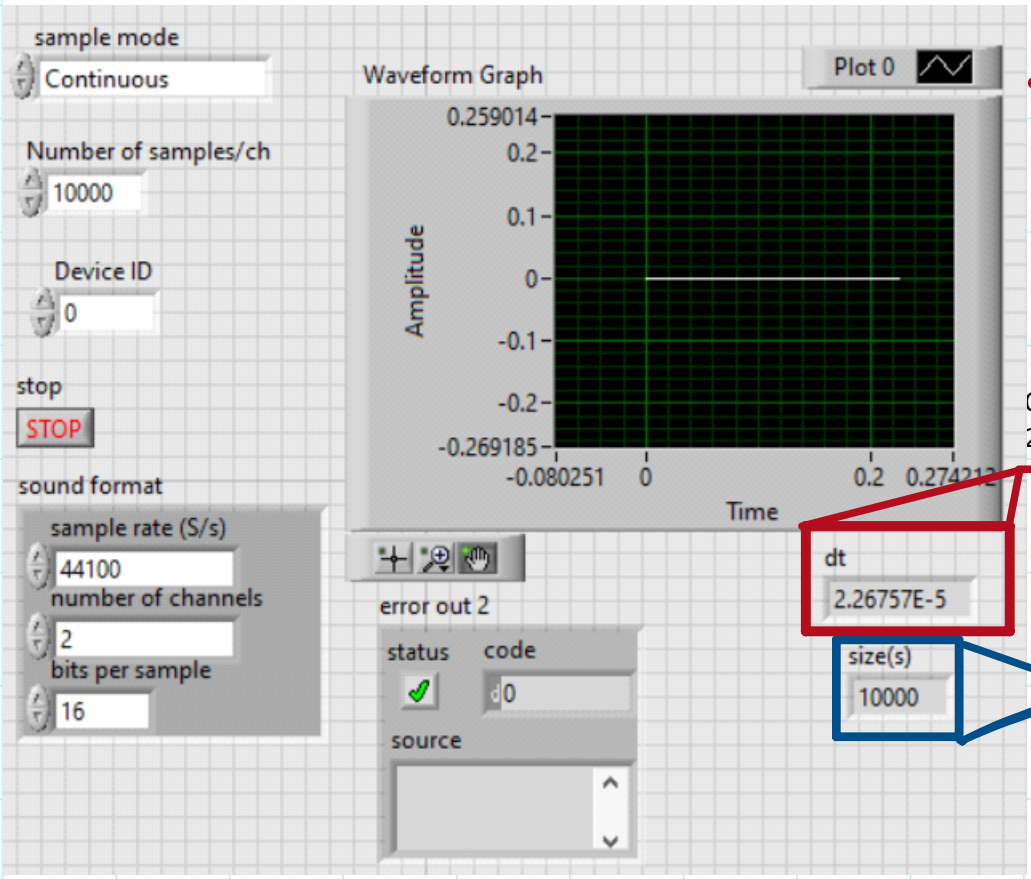
   $t = dt \times \text{number of samples/ch}$)

t= dt*numberOfSample/ch t = (1/44100)*(10000) = 0,2268

6. Now, find dt2 and size of the data array of **Sound Input Read VI** by using the following blocks.







Our dt output shows 2.26757E-5 that means 2.26757*10^(-5)=0.0000226757

And the size of the data array is 10000

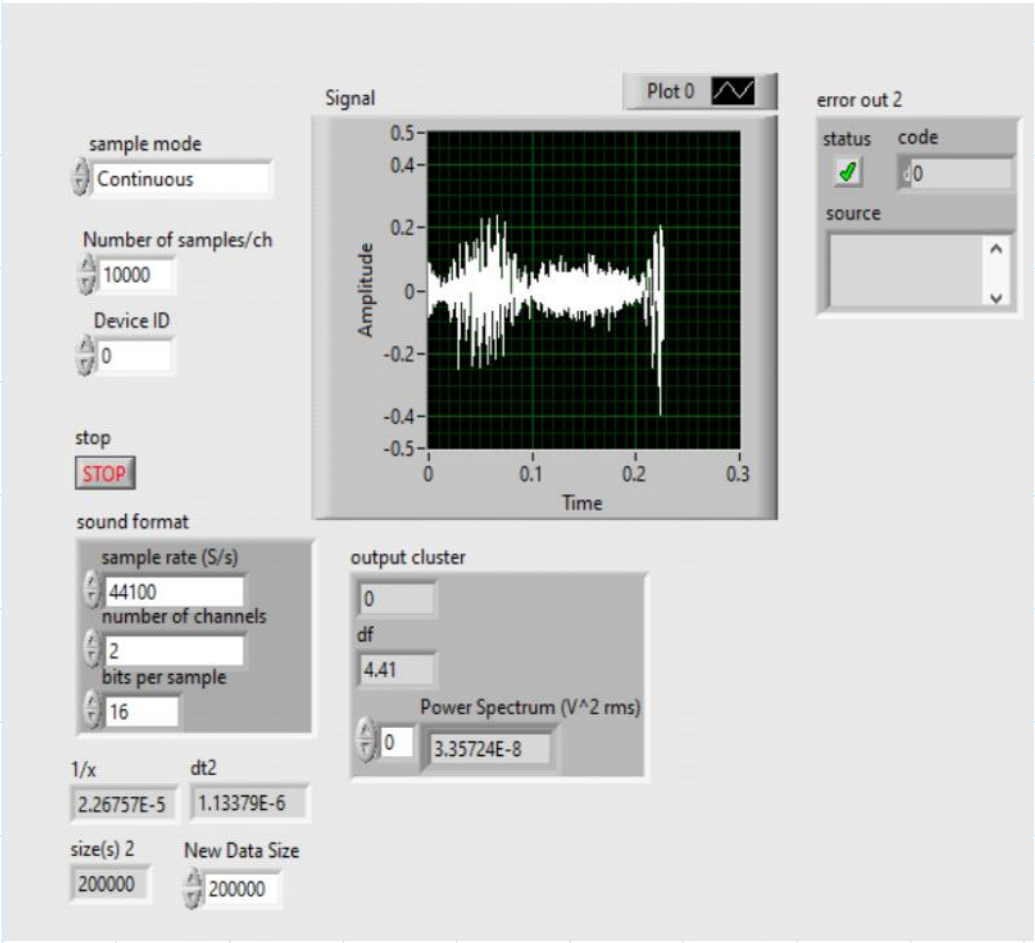
7. Compare dt in question 4 with dt2 in question 6. Also compare the size in question 6 with number of samples/ch.

In the Question 4 we have found dt = 1/44100=2.26757E-5 and at Question 6 our program showed us the dt is 2.26757E-5 we have calculated exactly same value

1. Compare the Size of data with "New Data Size" and number of samples/ch. Also compare dt with dt2. Comment on them.

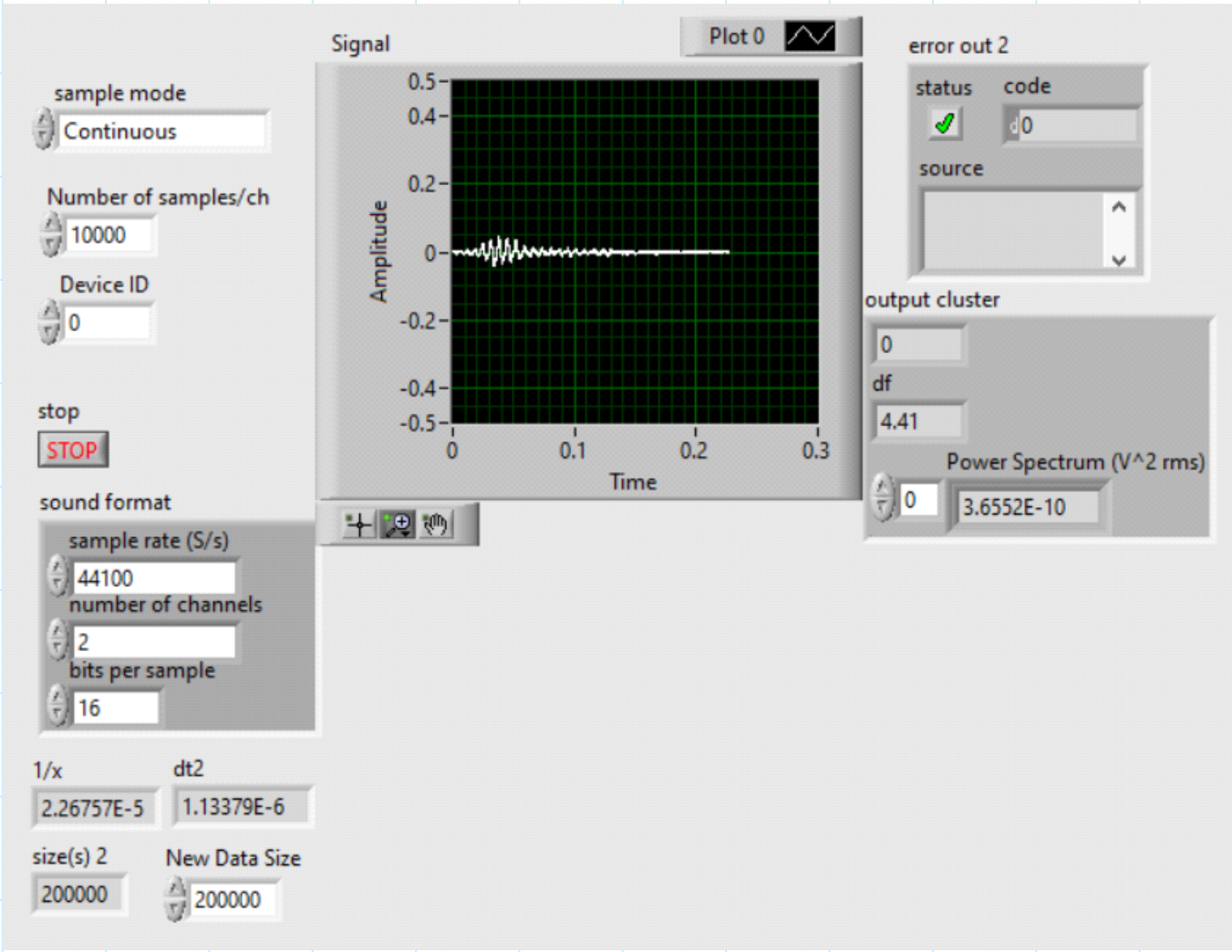
New data size is 200.000 and size of the data is 200.000 and number of samples/ch is 10.000 new data size and size of the data are equal and 20 times bigger than the number of samples/ch.
Value of the dt is 2.26757E-5 and dt2 is 1.13379E-6 dt is twice size of the dt2

2. Talk and see the spectrum of your voice. Stop the VI while talking and calculate the bandwidth of your voice.



Bandwith = SampleRate / 2
Bandwith = 44.100 / 2
Bandwith = 22050

3. Whistle gently and see the spectrum. Take snapshot of the spectrum and comment on your observation.



When we whistle, amplitude of the signal increases. And the power spectrum increases to 3.6552E-10 it was around E-12 at the start